Numerical Methods for PDEs

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Contents

1 Well-posed problems and reliable discretizations **1**

1.1 Why numerical methods? . . . 1

1.2 Discrete problems . . . 1

2 Elliptic tools: Green identities and maximum principles **4**

2.1 Gauss–Green formulas . . . 4

2.2 Harmonic functions . . . 4

2.3 Poisson's equation: uniqueness and stability . . . 4

2.4 A general second-order operator . . . 5

3 Centered finite differences in one dimension **7**

3.1 The model problem and its discretization . . . 7

3.2 The discrete maximum principle . . . 7

3.3 A sharp stability estimate . . . 8

3.4 Consistency and convergence . . . 8

3.5 Conditioning and solver cost . . . 9

4 Weak formulations and energy principles **10**

4.1 Why the strong formulation is too restrictive . . . 10

4.2 Sobolev spaces . . . 10

4.3 Derivation of the weak problem . . . 11

4.4 Dirichlet's principle . . . 11

5 Finite elements and finite volumes in one dimension **13**

5.1 The conforming Galerkin idea . . . 13

5.2 Well-posedness and error analysis . . . 13

5.3 When finite elements reproduce finite differences . . . 14

5.4 A finite volume derivation . . . 14

6 Variable coefficients and boundary conditions in one dimension **16**

6.1 A centered scheme . . . 16

6.2 Monotonicity and the discrete maximum principle . . . 16

6.3 Consistency . . . 17

6.4 Stability . . . 17

6.5 A Neumann boundary and ghost points . . . 18

7 Finite differences in several dimensions **20**

7.1 The five-point scheme 20

7.2 Discrete maximum principle and stability 20

7.3 Truncation error and convergence 21

7.4 General elliptic operators and mixed derivatives 21

7.5 A compact fourth-order Poisson scheme 21

8 Nonuniform grids and irregular boundaries **23**

8.1 The optimal three-point second derivative 23

8.2 What order is the nonuniform formula? 23

8.3 The Shortley–Weller idea 23

8.4 Why first-order local error need not destroy global order 24

9 Notation at a glance **25**

Suggested reading **25**

1 Well-posed problems and reliable discretizations

1.1 Why numerical methods?

Analytical theory answers structural questions: Does a solution exist? Is it unique? How does it depend on the data? Even when those questions have satisfactory answers, a closed formula is rarely available. Moreover, the solution belongs to an infinite-dimensional function space. A numerical method replaces the continuous problem by a finite-dimensional problem whose solution can be computed.

Write an abstract problem as

$$\mathcal{P}(u, g) = 0, \tag{1.1}$$

where g collects the forcing, coefficients, domain, and boundary data. When the solution is unique, the solution operator is denoted by $u = R(g)$.

Definition 1.1 (Hadamard well-posedness)**.** Problem (1.1) is well posed in specified data and solution norms if:

1. a solution exists for every admissible datum;
2. the solution is unique;
3. the solution depends continuously on the data.

In a locally Lipschitz setting, continuous dependence takes the form

$$\|R(g + \delta g) - R(g)\|_X \leq C_g \|\delta g\|_Y$$

for all sufficiently small admissible perturbations δg. For nonzero g and $R(g)$, a local relative condition number is

$$\kappa_R(g) = \limsup_{\varepsilon \downarrow 0} \sup_{0 < \|\delta g\|_Y \leq \varepsilon \|g\|_Y} \frac{\|R(g + \delta g) - R(g)\|_X \,/\, \|R(g)\|_X}{\|\delta g\|_Y \,/\, \|g\|_Y}.$$

A large condition number signals intrinsic sensitivity. *No numerical algorithm can remove sensitivity that already belongs to the continuous problem.*

1.2 Discrete problems

Let $h > 0$ denote a resolution parameter, usually a mesh width. For each h, let X_h be the finite-dimensional space of discrete unknowns and let Y_h be the space in which the discrete equations, residuals, and right-hand sides are measured. Both spaces are equipped with suitable norms. Once the discrete data g_h are fixed, the discrete operator acts as

$$\mathcal{P}_h(\,\cdot\,, g_h) : X_h \longrightarrow Y_h.$$

The numerical problem is

$$\mathcal{P}_h(u_h, g_h) = 0, \tag{1.2}$$

with $u_h \in X_h$. We first use the simplest notation, in which the exact solution can be inserted into the discrete equations and compared directly with u_h. The common situation in which sampling or interpolation is required is explained just after the basic error argument.

Definition 1.2 (Consistency)**.** Let u be a sufficiently smooth exact solution, and let g_h denote the discrete representation of the exact data. The residual obtained by substituting u into the scheme is the *truncation error*

$$\tau_h = \mathcal{P}_h(u, g_h) \in Y_h.$$

The method is consistent if

$$\|\tau_h\|_{Y_h} \longrightarrow 0 \qquad \text{as } h \to 0.$$

Definition 1.3 (Stability)**.** A family of discrete problems is stable if perturbations of the discrete data produce uniformly controlled perturbations of the discrete solution. For a linear operator $A_h : X_h \to Y_h$ and the problem $A_h u_h = f_h$, a standard form is

$$\left\|A_h^{-1}\right\|_{\mathcal{L}(Y_h, X_h)} \leq C \qquad \text{with } C \text{ independent of } h.$$

Definition 1.4 (Convergence)**.** A method is convergent if the numerical solution approaches the exact solution in the chosen solution norm:

$$\|u_h - u\|_{X_h} \longrightarrow 0 \qquad \text{as } h \to 0.$$

Proposition 1.5 (The basic error mechanism for linear problems)**.** *Suppose $A_h : X_h \to Y_h$, $A_h u_h = f_h$, and the exact solution can be inserted into the scheme so that $A_h u = f_h + \tau_h$. If $\left\|A_h^{-1}\right\| \leq C$ uniformly, then*

$$\|u - u_h\|_{X_h} \leq C \|\tau_h\|_{Y_h}.$$

Thus consistency plus uniform stability implies convergence.

Proof. The error $e_h = u - u_h$ satisfies $A_h e_h = \tau_h$, so $e_h = A_h^{-1}\tau_h$. □

Sampling, interpolation, and the approximation property

The preceding notation deliberately suppresses a technical point. In many methods, u is a function whereas an element of X_h is a vector of nodal values, cell averages, or coefficients. The expression $A_h u$ and the difference $u_h - u$ are then not literally defined.
Introduce a sampling or interpolation operator

$$I_h : X \longrightarrow X_h.$$

The precise versions of consistency and discrete convergence are then

$$\|A_h I_h u - f_h\|_{Y_h} \longrightarrow 0, \qquad \|u_h - I_h u\|_{X_h} \longrightarrow 0.$$

(Using the PDE point-wise gives $f_h = I_h A u$. Thus, consistency is a statement about the difference $A_h I_h u - I_h A u$.) Stability gives the second statement from the first exactly as above. To conclude that the numerical method approximates the function u, one also requires an *approximation property*. If $X_h \subset X$, a typical assumption is

$$\|u - I_h u\|_X \longrightarrow 0.$$

With compatible norms, the triangle inequality then gives

$$\|u - u_h\|_X \leq \|u - I_h u\|_X + \|I_h u - u_h\|_X \longrightarrow 0.$$

If X_h consists only of grid vectors, use a reconstruction $R_h : X_h \to X$. The corresponding approximation property is $\|u - R_h I_h u\|_X \to 0$; if R_h is uniformly bounded, then

$$\|u - R_h u_h\|_X \leq \|u - R_h I_h u\|_X + C_R \|I_h u - u_h\|_{X_h} \longrightarrow 0.$$

Thus, in the fully precise formulation, approximation, consistency, and stability together imply convergence to the continuous solution.

Scope of the Lax equivalence theorem

The slogan *consistency plus stability implies convergence* is often associated with the Lax–Richtmyer theorem. That theorem is classically formulated for linear, well-posed initial-value problems and should not be quoted as an unrestricted theorem for every PDE discretization. This is in contrast with the Dahlquist Equivalence Theorem on numerical methods for ODEs.

Exercises

Exercise 1.1 (Absolute versus relative conditioning)**.** Let $R(g) = g^{-1}$ for $g > 0$. Compute the absolute sensitivity $|R'(g)|$ and the relative condition number. Explain why the two quantities tell different stories near $g = 0$.

Exercise 1.2 (A scalar model)**.** Consider $a_h u_h = f_h$ with $a_h > 0$, $a_h \to a > 0$, and $f_h \to f$. Identify a consistency error and a stability constant, and prove $u_h \to f/a$.

Exercise 1.3 (Residual versus error)**.** Construct a 2×2 linear system for which a small residual does not imply a small error in the Euclidean norm. Relate the amplification to the condition number of the matrix.

Exercise 1.4 (Nonlinear extension)**.** Suppose $F_h : X_h \to X_h$ is differentiable near a discrete solution and $F_h'(u_h)^{-1}$ is uniformly bounded. Use a first-order Taylor expansion to formulate a local analogue of the basic error mechanism.

2 Elliptic tools: Green identities and maximum principles

2.1 Gauss–Green formulas

Let $\Omega \subset \mathbb{R}^d$ be bounded with sufficiently regular boundary, and let n denote the outward unit normal. For $u \in C^1(\overline{\Omega})$, the divergence theorem gives

$$\int_\Omega \partial_i u \, \mathrm{d}x = \int_{\partial\Omega} u \, n_i \, \mathrm{d}s, \qquad i = 1, \dots, d. \tag{2.1}$$

Consequently,

$$\int_\Omega \nabla u \, \mathrm{d}x = \int_{\partial\Omega} u \, n \, \mathrm{d}s, \tag{2.2}$$

$$\int_\Omega \Delta u \, \mathrm{d}x = \int_{\partial\Omega} \partial_n u \, \mathrm{d}s, \tag{2.3}$$

$$\int_\Omega \nabla u \cdot \nabla v \, \mathrm{d}x = -\int_\Omega v \, \Delta u \, \mathrm{d}x + \int_{\partial\Omega} v \, \partial_n u \, \mathrm{d}s. \tag{2.4}$$

The last identity is Green's first identity and is the bridge from strong to weak formulations.

2.2 Harmonic functions

A function $u \in C^2(\Omega)$ is harmonic if $\Delta u = 0$ in Ω. Harmonic functions satisfy the mean-value property:

$$u(x) = \fint_{\partial B_r(x)} u \, \mathrm{d}s = \fint_{B_r(x)} u \, \mathrm{d}y \qquad \text{whenever } \overline{B_r(x)} \subset \Omega.$$

Conversely, a continuous function with the mean-value property is harmonic.

Theorem 2.1 (Maximum principles for harmonic functions)**.** *Let Ω be connected and $u \in C^2(\Omega) \cap C(\overline{\Omega})$ be harmonic. Then*

$$\max_{\overline{\Omega}} u = \max_{\partial\Omega} u, \qquad \min_{\overline{\Omega}} u = \min_{\partial\Omega} u.$$

If an extremum is attained at an interior point, then u is constant.

The strong statement follows from the mean-value property: if $u(x_0)$ equals the global maximum, then the average of u over every small ball about x_0 can equal that maximum only when u is constant on the ball. Connectedness propagates the conclusion.

Corollary 2.2 (Positivity)**.** *If u is harmonic, $u = g \geq 0$ on $\partial\Omega$, and g is not identically zero, then $u > 0$ in Ω.*

2.3 Poisson's equation: uniqueness and stability

Consider

$$-\Delta u = f \quad \text{in } \Omega, \qquad u = g \quad \text{on } \partial\Omega. \tag{2.5}$$

Corollary 2.3 (Uniqueness)**.** *There is at most one classical solution of (2.5).*

Proof. The difference of two solutions is harmonic and vanishes on $\partial\Omega$; the maximum principle applied to w and $-w$ gives $w = 0$. □

An equivalent energy proof uses (2.4):

$$0 = -\int_\Omega w\Delta w \, \mathrm{d}x = \int_\Omega |\nabla w|^2 \, \mathrm{d}x,$$

because $w = 0$ on the boundary.

Theorem 2.4 (A maximum-norm a priori estimate)**.** *Suppose Ω is contained in a ball $B_R(x_0)$ and $u \in C^2(\Omega) \cap C(\overline{\Omega})$ solves (2.5). Then*

$$\|u\|_{L^\infty(\Omega)} \le \|g\|_{L^\infty(\partial\Omega)} + \frac{R^2}{2d} \|f\|_{L^\infty(\Omega)}.$$

Proof. Use the barrier $\phi(x) = (R^2 - |x - x_0|^2)/(2d)$, for which $-\Delta\phi = 1$ and $\phi \ge 0$ on Ω, and apply the maximum principle to suitable combinations of u, ϕ, and the boundary norm of g. □

If $u, \widetilde{u}$ correspond to data (f, g) and $(\widetilde{f}, \widetilde{g})$, the same estimate gives

$$\|u - \widetilde{u}\|_{L^\infty(\Omega)} \le \|g - \widetilde{g}\|_{L^\infty(\partial\Omega)} + \frac{R^2}{2d} \left\|f - \widetilde{f}\right\|_{L^\infty(\Omega)}.$$

This is continuous dependence, hence the stability part of well-posedness.

2.4 A general second-order operator

For clarity, write a linear operator in nondivergence form. Let $\Omega \subset \mathbb{R}^d$ be bounded and let

$$\mathcal{L}u = -\sum_{i,j=1}^{d} a_{ij}(x)\partial_{ij}u + \sum_{i=1}^{d} b_i(x)\partial_i u + c(x)u.$$

Assume that $A = (a_{ij})$ is symmetric and uniformly elliptic:

$$\lambda |\xi|^2 \le \xi^\top A(x)\xi \le \Lambda |\xi|^2, \qquad \lambda > 0.$$

Theorem 2.5 (Weak maximum and comparison principles)**.** *If $a_{ij}, b_i, c \in C(\overline{\Omega})$, with $c \ge 0$, then, for every $u \in C^2(\Omega) \cap C(\overline{\Omega})$:*

1. *if $\mathcal{L}u \le 0$ in Ω, then*

$$\max_{\overline{\Omega}} u \le \max\left\{0, \max_{\partial\Omega} u\right\};$$

   if $c \equiv 0$, the zero may be omitted;

2. *if $\mathcal{L}u \le \mathcal{L}v$ in Ω and $u \le v$ on $\partial\Omega$, then $u \le v$ in $\overline{\Omega}$.*

Remark 2.6. Analogous results hold for divergence-form operators $-\nabla \cdot (A\nabla u) + b \cdot \nabla u + cu$, but the natural weak hypotheses and proof are formulated in Sobolev spaces.

Exercises

Exercise 2.1 (Green's second identity)**.** Derive

$$\int_\Omega (u\Delta v - v\Delta u) \, \mathrm{d}x = \int_{\partial\Omega} (u\partial_n v - v\partial_n u) \, \mathrm{d}s.$$

Exercise 2.2 (Maximum-principle uniqueness)**.** Let $c \geq 0$. Prove uniqueness for $\mathcal{L}u = f$ with prescribed Dirichlet data by applying the comparison principle to the difference of two solutions.

Exercise 2.3 (Constructing a barrier)**.** For $\Omega = (0,1)^d$, construct a quadratic $\phi \geq 0$ satisfying $-\Delta\phi = 1$. Compare the resulting stability constant with the ball-based constant above.

Exercise 2.4 (Why $c \geq 0$?)**.** Give a one-dimensional example showing that the maximum principle can fail when the zeroth-order coefficient is sufficiently negative.

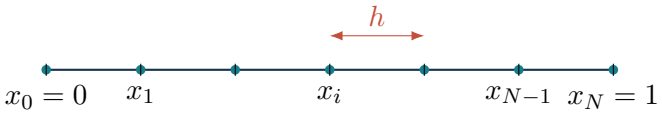
3 Centered finite differences in one dimension

3.1 The model problem and its discretization

Consider

$$-u'' = f \quad \text{in } (0,1), \qquad u(0) = u(1) = 0. \tag{3.1}$$

Choose $N \in \mathbb{N}$, set $h = 1/N$, and define $x_i = ih$, $i = 0, \dots, N$.



The centered second difference is

$$\delta_h^2 v(x) = \frac{v(x+h) - 2v(x) + v(x-h)}{h^2}.$$

The finite-difference scheme seeks nodal values $U_i \approx u(x_i)$ such that

$$(\mathcal{L}_h U)_i := -\delta_h^2 U_i = f_i := f(x_i), \quad i = 1, \dots, N-1, \qquad U_0 = U_N = 0. \tag{3.2}$$

In matrix form,

$$A_h \mathbf{U} = \mathbf{F}, \qquad A_h = \frac{1}{h^2} \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{pmatrix}.$$

The matrix is symmetric positive definite.

3.2 The discrete maximum principle

Theorem 3.1 (Discrete maximum principle)**.** *Let $V = (V_0, \dots, V_N)$. If $(\mathcal{L}_h V)_i \le 0$ for $i = 1, \dots, N-1$, then*

$$\max_{0 \le i \le N} V_i = \max\{V_0, V_N\}.$$

If an interior maximum is attained and $\mathcal{L}_h V \le 0$, then the maximum propagates through neighboring nodes; hence V is constant.

Proof. At an interior maximizer n,

$$0 \le -h^2 (\mathcal{L}_h V)_n = V_{n+1} - 2V_n + V_{n-1},$$

whereas maximality gives $(V_{n+1} + V_{n-1})/2 \le V_n$. Equality is forced, so both neighbors also attain the maximum. Repetition reaches the boundary. □

3.3 A sharp stability estimate

Set $w(x) = x(1-x)$ and $W_i = w(x_i)$. Since w is quadratic,

$$\mathcal{L}_h W_i = -w''(x_i) = 2, \qquad W_0 = W_N = 0, \qquad 0 \le W_i \le \frac{1}{4}.$$

For an arbitrary grid function V, apply the discrete maximum principle to

$$Z_i^{\pm} = \pm V_i - \frac{1}{2}\|\mathcal{L}_h V\|_{\ell^\infty} W_i.$$

This yields:

Lemma 3.2 (Maximum-norm stability)**.** *For every grid function V,*

$$\|V\|_{\ell^\infty} \le \max\{|V_0|, |V_N|\} + \frac{1}{8}\|\mathcal{L}_h V\|_{\ell^\infty}. \tag{3.3}$$

In particular, the homogeneous discrete problem has only the zero solution. Since the system is square and finite dimensional, uniqueness also implies existence.

3.4 Consistency and convergence

For $u \in C^4([0,1])$, Taylor's theorem with remainder gives

$$u''(x) - \delta_h^2 u(x) = -\frac{h^2}{24}\left(u^{(4)}(\xi_-) + u^{(4)}(\xi_+)\right), \tag{3.4}$$

where $\xi_- \in (x-h, x)$ and $\xi_+ \in (x, x+h)$. Therefore

$$\left|u''(x) - \delta_h^2 u(x)\right| \le \frac{h^2}{12}\left\|u^{(4)}\right\|_{L^\infty(0,1)}.$$

Note that the remainder in (3.4) carries a minus sign when written as $u'' - \delta_h^2 u$. The sign does not affect the norm bound, but it matters in a pointwise asymptotic expansion.

Let $e_i = u(x_i) - U_i$. Since $e_0 = e_N = 0$,

$$(\mathcal{L}_h e)_i = (\mathcal{L}_h I_h u)_i - f_i = u''(x_i) - \delta_h^2 u(x_i).$$

Combining the truncation bound with (3.3) proves:

Theorem 3.3 (Second-order maximum-norm convergence)**.** *If $u \in C^4([0,1])$ solves (3.1) and U solves (3.2), then*

$$\max_{0 \le i \le N} |u(x_i) - U_i| \le \frac{h^2}{96}\left\|u^{(4)}\right\|_{L^\infty(0,1)}. \tag{3.5}$$

The reusable proof pattern

error equation + uniform stability
+ truncation estimate $\Longrightarrow$ convergence estimate.

3.5 Conditioning and solver cost

Spectral estimate for the Poisson matrix

The eigenvalues of A_h are

$$\lambda_k = \frac{4}{h^2}\sin^2\left(\frac{k\pi}{2N}\right), \qquad k = 1, \dots, N-1.$$

Hence

$$\operatorname{cond}_2(A_h) \sim \frac{4}{\pi^2 h^2}.$$

Refinement improves approximation but worsens algebraic conditioning. This observation motivates preconditioners and multigrid methods.

Exercises

Exercise 3.1 (Exactness on quadratics)**.** Prove that $\delta_h^2 p = p''$ for every polynomial of degree at most three. Find the error explicitly for $p(x) = x^4$.

Exercise 3.2 (A nonzero boundary condition)**.** Derive the matrix system for $u(0) = \alpha$, $u(1) = \beta$. Show how the boundary values enter only the first and last entries of the right-hand side.

Exercise 3.3 (Energy stability)**.** Multiply $(\mathcal{L}_h U)_i = f_i$ by hU_i, sum over the interior nodes, and prove a discrete integration-by-parts identity. Deduce a stability estimate in the discrete H^1 seminorm.

Exercise 3.4 (Observed order)**.** Use the exact solution $u(x) = \sin(\pi x)$. Compute the discrete solution for successively halved h, and verify that $\|I_h u - U\|_{\ell^\infty}$ decreases by a factor close to four.

4 Weak formulations and energy principles

4.1 Why the strong formulation is too restrictive

The classical problem $-u'' = f$ appears to require two derivatives of u and pointwise values of f. Many physically natural loads do not produce such smooth solutions.

Example 4.1 (A discontinuous load)**.** If $f = \chi_{[0.4,0.6]}$, then the solution of $-u'' = f$, $u(0) = u(1) = 0$, is positive and piecewise quadratic. It is C^1, but its second derivative jumps at 0.4 and 0.6.

Example 4.2 (A point load)**.** Let $f = -\delta_{1/2}$. The weak solution is the negative tent function

$$u(x) = \begin{cases} -x/2, & 0 \le x \le 1/2, \\ -(1-x)/2, & 1/2 \le x \le 1. \end{cases}$$

It belongs to $H_0^1(0,1)$, but it is not classically differentiable at $x = 1/2$.

With the convention $-u'' = f$, a nonnegative transverse load produces a nonnegative displacement for homogeneous Dirichlet data. A negative tent corresponds to the negative point load $-\delta_{1/2}$, not to a positive load.

4.2 Sobolev spaces

Define

$$L^2(0,1) = \left\{ v : \|v\|_{L^2}^2 = \int_0^1 |v|^2 \, \mathrm{d}x < \infty \right\},$$

$$H^1(0,1) = \left\{ v \in L^2(0,1) : v' \in L^2(0,1) \right\}, \qquad H_0^1(0,1) = \overline{C_c^\infty(0,1)}^{H^1}.$$

In one dimension, functions in H^1 have continuous representatives, and

$$H_0^1(0,1) = \left\{ v \in H^1(0,1) : v(0) = v(1) = 0 \right\}.$$

Poincaré's inequality gives

$$\|v\|_{L^2(0,1)} \le C_P \, \|v'\|_{L^2(0,1)} \qquad \forall v \in H_0^1(0,1).$$

Thus

$$\|v\|_V := \|v'\|_{L^2(0,1)}$$

is a norm on $V = H_0^1(0,1)$, equivalent to the full H^1 norm. The dual space $V' = H^{-1}(0,1)$ contains $L^2(0,1)$ and, in one dimension, point loads.

Norm versus seminorm

The quantity $\|v'\|_{L^2}$ is only a seminorm on $H^1(0,1)$, because constants have zero derivative. It becomes a genuine norm on $H_0^1(0,1)$ by Poincaré's inequality.

4.3 Derivation of the weak problem

Multiply $-u'' = f$ by a test function v satisfying $v(0) = v(1) = 0$, integrate, and use integration by parts:

$$-\int_0^1 u''v \, \mathrm{d}x = \int_0^1 u'v' \, \mathrm{d}x = \langle f, v \rangle_{V',V} .$$

The weak formulation is:

$$\text{Find } u \in V \text{ such that} \qquad a(u,v) = F(v) \quad \forall v \in V, \tag{4.1}$$

with

$$a(u,v) = \int_0^1 u'v' \, \mathrm{d}x, \qquad F(v) = \langle f, v \rangle_{V',V} .$$

Theorem 4.3 (Well-posedness of the weak problem)**.** *For every $F \in V'$, problem (4.1) has a unique solution, and*

$$\|u\|_V \le \|F\|_{V'} .$$

Proof. The bilinear form is continuous and coercive:

$$|a(w,v)| \le \|w\|_V \, \|v\|_V , \qquad a(v,v) = \|v\|_V^2 .$$

Hence, the Lax–Milgram Lemma (see later) gives existence and uniqueness. Testing with $v = u$ gives $\|u\|_V^2 = F(u) \le \|F\|_{V'} \, \|u\|_V$. □

4.4 Dirichlet's principle

Define the potential energy

$$J(v) = \frac{1}{2} a(v,v) - F(v) = \frac{1}{2} \int_0^1 |v'|^2 \, \mathrm{d}x - F(v). \tag{4.2}$$

Theorem 4.4 (Weak solution equals energy minimizer)**.** *The weak solution u of (4.1) is the unique minimizer of J over V.*

Proof. For $w \in V$,

$$J(u+w) = J(u) + a(u,w) - F(w) + \frac{1}{2} a(w,w) = J(u) + \frac{1}{2} a(w,w) \ge J(u).$$

Conversely, if u minimizes J, then the derivative of $t \mapsto J(u + tw)$ at $t = 0$ vanishes for every $w \in V$, which gives $a(u,w) = F(w)$. □

Nonhomogeneous Dirichlet data

If $u(0) = \alpha$ and $u(1) = \beta$, choose a lifting $G \in H^1(0,1)$ with the prescribed traces and write $u = G + w$, $w \in H_0^1(0,1)$. The admissible set is the affine space $G + H_0^1(0,1)$; variations still belong to $H_0^1(0,1)$.

Exercises

Exercise 4.1 (Poincaré inequality in one dimension)**.** Use $v(x) = \int_0^x v'(s)\,\mathrm{d}s$ and Cauchy–Schwarz to prove a Poincaré inequality on $H_0^1(0,1)$. Find a valid constant.

Exercise 4.2 (Point evaluation)**.** Prove that $|v(1/2)| \le C\,\|v'\|_{L^2}$ for $v \in H_0^1(0,1)$. Conclude that $\delta_{1/2} \in H^{-1}(0,1)$.

Exercise 4.3 (Energy of the tent function)**.** Verify distributionally that the tent function in the point-load example solves $-u'' = -\delta_{1/2}$. Compute $J(u)$.

Exercise 4.4 (Reaction term)**.** Derive the weak and energy formulations for $-u'' + cu = f$, $c \ge 0$. Identify the coercivity constant in the H_0^1 norm used above.

5 Finite elements and finite volumes in one dimension

5.1 The conforming Galerkin idea

The finite element method restricts the weak problem to a finite-dimensional subspace $V_h \subset V$:

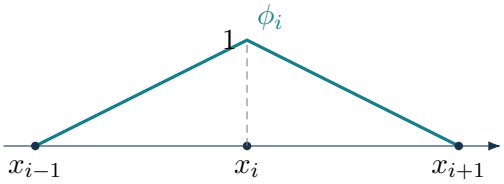
$$\text{Find } u_h \in V_h \text{ such that} \qquad a(u_h, v_h) = F(v_h) \quad \forall v_h \in V_h. \tag{5.1}$$

On the grid from section 3, choose the continuous, piecewise affine space

$$V_h = \left\{ v_h \in H_0^1(0,1) : v_h|_{[x_{i-1},x_i]} \in \mathbb{P}_1 \text{ for } i = 1, \dots, N \right\}.$$

Its nodal basis consists of hat functions ϕ_i, $i = 1, \dots, N-1$, satisfying $\phi_i(x_j) = \delta_{ij}$:

$$\phi_i(x) = \begin{cases} (x - x_{i-1})/h, & x \in [x_{i-1}, x_i], \\ (x_{i+1} - x)/h, & x \in [x_i, x_{i+1}], \\ 0, & \text{otherwise.} \end{cases}$$



Writing

$$u_h = \sum_{j=1}^{N-1} U_j \phi_j$$

and testing with every ϕ_i gives

$$K\mathbf{U} = \mathbf{b}, \qquad K_{ij} = \int_0^1 \phi_j' \phi_i' \, \mathrm{d}x, \qquad b_i = F(\phi_i). \tag{5.2}$$

For a uniform grid,

$$K = \frac{1}{h} \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{pmatrix}.$$

5.2 Well-posedness and error analysis

Because $V_h \subset V$, the same continuity and coercivity constants apply to the discrete problem. Hence (5.1) has a unique solution. Subtracting the continuous and discrete equations gives *Galerkin orthogonality*:

$$a(u - u_h, v_h) = 0 \qquad \forall v_h \in V_h. \tag{5.3}$$

Theorem 5.1 (Céa's lemma)**.** *If a is continuous with constant M and coercive with constant α, then*

$$\|u - u_h\|_V \le \frac{M}{\alpha} \inf_{v_h \in V_h} \|u - v_h\|_V .$$

Proof. For any $v_h \in V_h$, coercivity and (5.3) give

$$\alpha \|u - u_h\|_V^2 \le a(u - u_h, u - u_h) = a(u - u_h, u - v_h) \le M \|u - u_h\|_V \|u - v_h\|_V .$$

□

For piecewise affine elements and $u \in H^2(0,1)$, interpolation yields

$$\inf_{v_h \in V_h} \|u - v_h\|_{H^1} \le Ch \|u\|_{H^2} .$$

Thus the energy error is first order. A duality argument gives a second-order L^2 estimate under the corresponding elliptic regularity.

5.3 When finite elements reproduce finite differences

For $F(v) = \int_0^1 fv \, \mathrm{d}x$, the i-th finite element equation is

$$-\frac{1}{h}U_{i-1} + \frac{2}{h}U_i - \frac{1}{h}U_{i+1} = \int_{x_{i-1}}^{x_{i+1}} f\phi_i \, \mathrm{d}x.$$

Approximating the right-hand side by the nodal quadrature

$$\int_{x_{i-1}}^{x_{i+1}} f\phi_i \, \mathrm{d}x \approx hf(x_i)$$

and dividing by h recovers the centered finite-difference equation.

The equivalence uses quadrature

The linear finite element stiffness matrix and the centered finite-difference matrix have the same tridiagonal pattern, up to a row scaling. Exact equivalence for a general load requires a specific approximation of the finite element load vector; it is not an identity with exact integration.

5.4 A finite volume derivation

Let the control volume about x_i be $I_i = [x_{i-1/2}, x_{i+1/2}]$. Integrating $-u'' = f$ over I_i gives the exact balance

$$-u'(x_{i+1/2}) + u'(x_{i-1/2}) = \int_{I_i} f \, \mathrm{d}x. \tag{5.4}$$

Approximate the face fluxes by

$$u'(x_{i+1/2}) \approx \frac{U_{i+1} - U_i}{h}, \qquad u'(x_{i-1/2}) \approx \frac{U_i - U_{i-1}}{h},$$

and use the midpoint rule on the right. After division by h,

$$\frac{-U_{i+1} + 2U_i - U_{i-1}}{h^2} = f(x_i).$$

Thus the same algebraic scheme arises from three perspectives: pointwise differentiation, a variational principle with nodal quadrature, and local flux balance.

By construction, the finite volume approach allows to show a local conservation property as follows. Write the balance on cell i as

$$-q_{i+1/2} + q_{i-1/2} = \int_{I_i} f \, \mathrm{d}x, \qquad q_{i+1/2} \approx \frac{U_{i+1} - U_i}{h}.$$

Summing the finite volume equations over a consecutive block of control volumes, that is from $i = m$ to $i = n$, with $m < n$, gives

$$\sum_{i=m}^{n} (-q_{i+1/2} + q_{i-1/2}) = q_{m-1/2} - q_{n+1/2} = \int_{x_{m-1/2}}^{x_{n+1/2}} f \, \mathrm{d}x.$$

Every internal face flux appears once with each sign and cancels. The remaining equation says that the net outward flux through the boundary of the block equals the total source inside it. This is local conservation on any union of control volumes.

How the viewpoints differ

- Finite differences make truncation and maximum-principle arguments transparent.
- Finite elements naturally handle weak solutions, complex geometry, and energy-norm error estimates.
- Finite volumes enforce a local conservation law and are often preferred when flux balance is central.

The algebraic coincidence in this one-dimensional example should not be mistaken for complete equivalence of the methods.

Exercises

Exercise 5.1 (Element assembly)**.** Compute the 2×2 stiffness matrix on a single interval $[x_{i-1}, x_i]$, then assemble the global tridiagonal matrix.

Exercise 5.2 (Galerkin best approximation)**.** For the Poisson bilinear form, show that u_h is the orthogonal projection of u onto V_h in the energy inner product. Improve Céa's estimate to an equality involving the best approximation error.

Exercise 5.3 (A point load in FEM)**.** Take $F(v) = -v(1/2)$ on a mesh containing $1/2$. Write the finite element load vector and show that the exact tent solution belongs to V_h. What does the discrete solution equal?

6 Variable coefficients and boundary conditions in one dimension

6.1 A centered scheme

Consider the nondivergence-form boundary-value problem

$$\mathcal{L}u := -a(x)u'' + b(x)u' + c(x)u = f(x), \qquad u(0) = u_0, \quad u(1) = u_1, \tag{6.1}$$

where $a(x) \geq a_0 > 0$ and $c(x) \geq 0$. At an interior node define

$$(\mathcal{L}_h U)_i = -a_i \frac{U_{i+1} - 2U_i + U_{i-1}}{h^2} + b_i \frac{U_{i+1} - U_{i-1}}{2h} + c_i U_i. \tag{6.2}$$

The lower, diagonal, and upper coefficients in row i are

$$-\frac{a_i}{h^2} - \frac{b_i}{2h}, \qquad \frac{2a_i}{h^2} + c_i, \qquad -\frac{a_i}{h^2} + \frac{b_i}{2h}.$$

6.2 Monotonicity and the discrete maximum principle

The off-diagonal coefficients are nonpositive provided

$$a_i - \frac{h}{2}|b_i| \geq 0. \tag{6.3}$$

This is a mesh Péclet restriction: centered convection must not overwhelm diffusion at the grid scale (more in the following sections).

Theorem 6.1 (Discrete comparison principle)**.** *Suppose (6.3) holds. If $(\mathcal{L}_h V)_i \leq 0$ at all interior nodes, then*

$$\max_i V_i \leq \begin{cases} \max\{V_0, V_N\}, & c \equiv 0, \\ \max\{V_0, V_N, 0\}, & c \geq 0. \end{cases}$$

Consequently, if $\mathcal{L}_h U \leq \mathcal{L}_h W$ in the interior and $U \leq W$ on the boundary, then $U \leq W$ everywhere.

Let A_h denote the interior matrix corresponding to homogeneous Dirichlet boundary values. Then A_h has a positive diagonal and nonpositive off-diagonal entries. Its inverse is nonnegative. This is the characteristic structure of a nonsingular M-matrix[1]. The comparison principle shows that the M-matrix property is the algebraic form of monotonicity.

The discrete maximum principle first implies that A_h is nonsingular: if $A_h V = 0$, applying the principle to V and $-V$ gives $V = 0$. If $F \geq 0$ and $A_h V = F$, then $\mathcal{L}_h(-V) = -F \leq 0$, with zero boundary values. Hence $-V \leq 0$, so $V \geq 0$. Thus A_h^{-1} maps nonnegative vectors to nonnegative vectors. Applying this observation to each coordinate vector shows that $A_h^{-1} \geq 0$ componentwise. Consequently, A_h is a nonsingular M-matrix.

[1] A matrix A is an M-matrix if its off-diagonal etries satisfy $a_{ij} \leq 0$ and its inverse A^{-1} exists with $A^{-1} \geq 0$. The diagonal entries of an M-matrix are positive; that is, $a_{ii} > 0$. An equivalent condition for M-matrix (M-criterion) is the following. Assume $a_{ij} \leq 0$, $i \neq j$. Then, A is an M-matrix if and only if there exists a vector $e > 0$ such that $Ae > 0$. The barrier function used above in Section 3.3 and again below in Section 6.4 has precisely the meaning of providing such vector. IMoreover, constructing the barrier not only provides an alternative M-matrix proof but also yields a stability constant.

6.3 Consistency

For a smooth u, the centered first derivative satisfies

$$\frac{u(x+h)-u(x-h)}{2h} = u'(x) + \frac{h^2}{6}u^{(3)}(\eta)$$

for some $\eta \in (x-h, x+h)$. Combining this with (3.4) gives

$$|(\mathcal{L}_h I_h u)_i - (\mathcal{L}u)(x_i)| \le h^2 \left(\frac{\|a\|_{L^\infty}}{12} \left\|u^{(4)}\right\|_{L^\infty} + \frac{\|b\|_{L^\infty}}{6} \left\|u^{(3)}\right\|_{L^\infty} \right).$$

Note, in particular, that the reaction term $c\,u$ term is sampled exactly and thus it does not appear in the consistency estimate.

6.4 Stability

When $b = 0$, we can use again the barrier function $w = x(1-x)$ which obeys

$$(\mathcal{L}_h w)_i = 2a_i + c_i w_i \ge 2a_0.$$

The comparison argument therefore yields

$$\|V\|_{\ell^\infty} \le \max\{|V_0|, |V_N|\} + \frac{1}{8a_0} \|\mathcal{L}_h V\|_{\ell^\infty}.$$

With $b \neq 0$, things are more complicated, and a different barrier function needs to be used. Still maximum-norm stability can be obtained with a stability constant depending on a_0, b, c, and the interval, provided the monotonicity condition holds.

Theorem 6.2 (Maximum-norm stability)**.** *Assume $a_i \ge a_0 > 0$, $c_i \ge 0$, and $|b_i| \le B$, and suppose the mesh Péclet condition (6.3) holds. Then there are constants $h_0 > 0$ and $C_{\mathrm{stab}} > 0$, independent of h, such that every grid function V satisfies, for $h \le h_0$,*

$$\|V\|_{\ell^\infty} \le \max\{|V_0|, |V_N|\} + C_{\mathrm{stab}} \|\mathcal{L}_h V\|_{\ell^\infty}. \tag{6.4}$$

As before, once consistency and stability are ensured, convergence follows.

Theorem 6.3 (Second-order maximum-norm convergence)**.** *Let $u \in C^4([0,1])$ solve (6.1), and let U solve the centered scheme (6.2) with the exact Dirichlet boundary values. Under the hypotheses of the maximum-norm stability theorem, for all sufficiently small h,*

$$\|I_h u - U\|_{\ell^\infty} \le C_{\mathrm{stab}} h^2 \left(\frac{\|a\|_{L^\infty}}{12} \left\|u^{(4)}\right\|_{L^\infty} + \frac{\|b\|_{L^\infty}}{6} \left\|u^{(3)}\right\|_{L^\infty} \right). \tag{6.5}$$

In particular, the centered scheme converges with order two in the nodal maximum norm.

Proof. Set $e = I_h u - U$. The boundary values are exact, so $e_0 = e_N = 0$, and the error equation is

$$\mathcal{L}_h e = \mathcal{L}_h I_h u - I_h(\mathcal{L}u).$$

Apply (6.4) and then the preceding truncation estimate. □

Divergence form and conservative fluxes

For

$$-(a(x)u')' + c(x)u = f,$$

it is generally preferable to approximate the face flux $a\,u'$:

$$-\frac{1}{h}\left[a_{i+1/2}\frac{U_{i+1}-U_i}{h} - a_{i-1/2}\frac{U_i - U_{i-1}}{h}\right] + c_i U_i = f_i.$$

This discretization is conservative and corresponds directly to a finite volume balance (see below). It is not the same as simply sampling $-a(x)u''$.

6.5 A Neumann boundary and ghost points

Suppose the right boundary condition is $u'(1) = q$. A centered ghost relation is

$$\frac{U_{N+1} - U_{N-1}}{2h} = q.$$

For $q = 0$, substituting $U_{N+1} = U_{N-1}$ into the differential equation at x_N gives, when b_N drops out,

$$2a_N\frac{U_N - U_{N-1}}{h^2} + c_N U_N = f_N.$$

Boundary accuracy must be checked separately

The centered ghost relation is second order as an approximation to the derivative, but the eliminated boundary equation above has only first-order local truncation error in general when written in the same PDE scaling as the interior equations. Nevertheless, the ghost-point method may still be globally second-order accurate because this lower-order residual occurs only at the boundary and its influence is weakened by the discrete solution operator. Establishing second-order convergence therefore requires a boundary-sensitive stability argument, a suitably scaled boundary equation, or a discrete Green's-function estimate; it does not follow directly from the maximum norm of the local truncation errors. An alternative treatment of the Neumann condition which has a second-order truncation error can be obtained by exploiting directly the following second-order one-sided divided difference:

$$\frac{3U_N - 4U_{N-1} + U_{N-2}}{2h} = q.$$

Exercises

Exercise 6.1 (Matrix coefficients)**.** Derive the three coefficients of (6.2). Show that their row sum is c_i.

Exercise 6.2 (Mesh Péclet number)**.** For constant $a > 0$ and b, write (6.3) as a condition on $\mathrm{Pe}_h = |b|\,h/(2a)$. Numerically compare centered and first-order upwind differences when $\mathrm{Pe}_h > 1$.

Exercise 6.3 (A conservative interface problem)**.** Let a be piecewise constant with a jump at $x = 1/2$. Derive a finite volume scheme for $-(au')' = f$. Explain why the face value of a should be a harmonic rather than arithmetic average when the interface lies between nodes.

Exercise 6.4 (Neumann nullspace)**.** Consider $-u'' = f$ with $u'(0) = u'(1) = 0$. Find the compatibility condition on f, identify the continuous and discrete nullspaces, and state one normalization that restores uniqueness.

Exercise 6.5 (Second-order accuracy of the ghost-point method)**.** Consider the mixed boundary-value problem with right boundary condition $u'(1) = q$ discretised using the ghost point method. Show that the local truncation error at the right endpoint is only $O(h)$, whereas the interior truncation errors are $O(h^2)$. Prove nevertheless that

$$\max_{0 \le i \le N} |U_i - u(x_i)| \le Ch^2,$$

where C is independent of h.

7 Finite differences in several dimensions

7.1 The five-point scheme

Consider

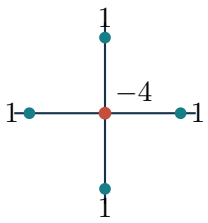
$$-\Delta u = f \quad \text{in } \Omega = (0,1)^2, \qquad u = 0 \quad \text{on } \partial\Omega. \tag{7.1}$$

On the uniform grid $x_i = ih$, $y_j = jh$, $h = 1/N$, define

$$\Delta_h U_{ij} = \frac{U_{i+1,j} + U_{i-1,j} + U_{i,j+1} + U_{i,j-1} - 4U_{ij}}{h^2}.$$

The five-point method is

$$-\Delta_h U_{ij} = f(x_i, y_j) \quad (1 \le i, j \le N-1), \qquad U_{ij} = 0 \quad \text{on boundary nodes.} \tag{7.2}$$



With lexicographic ordering of the $(N-1)^2$ unknowns,

$$A_h = \frac{1}{h^2}(T \otimes I + I \otimes T), \qquad T = \begin{pmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & \ddots & \ddots & \ddots \\ & & -1 & 2 \end{pmatrix}.$$

The matrix is sparse, symmetric positive definite, and block tridiagonal.

7.2 Discrete maximum principle and stability

Theorem 7.1 (Two-dimensional discrete maximum principle)**.** *If $-\Delta_h V_{ij} \le 0$ at every interior node, then the maximum of V is attained on the grid boundary.*

Proof. At an interior maximizer,

$$V_{ij} \le \frac{1}{4}\left(V_{i+1,j} + V_{i-1,j} + V_{i,j+1} + V_{i,j-1}\right) \le V_{ij}.$$

Equality forces all four neighbors to share the maximum. Connectivity propagates it to the boundary. □

Use the discrete barrier

$$w(x,y) = \frac{x(1-x) + y(1-y)}{4}.$$

It satisfies $-\Delta_h w = 1$, vanishes only at the four corners but is nonnegative on the whole boundary, and $\max_{\overline{\Omega}} w = 1/8$. Comparison gives

$$\|V\|_{\ell^\infty(\Omega_h)} \le \|V\|_{\ell^\infty(\partial\Omega_h)} + \frac{1}{8}\,\|-\Delta_h V\|_{\ell^\infty(\Omega_h^\circ)}\,. \tag{7.3}$$

7.3 Truncation error and convergence

Let

$$M_{4,0} = \left\|\partial_x^4 u\right\|_{L^\infty(\Omega)}, \qquad M_{0,4} = \left\|\partial_y^4 u\right\|_{L^\infty(\Omega)}.$$

Applying the one-dimensional centered-difference identity in each coordinate gives

$$\|(-\Delta_h I_h u) - f\|_{\ell^\infty} \le \frac{h^2}{12}(M_{4,0} + M_{0,4}).$$

The error equation and (7.3) yield:

Theorem 7.2 (Five-point convergence)**.** *If $u \in C^4(\overline{\Omega})$ solves (7.1), then the solution of (7.2) satisfies*

$$\max_{i,j} |u(x_i, y_j) - U_{ij}| \le \frac{h^2}{96}(M_{4,0} + M_{0,4}). \tag{7.4}$$

7.4 General elliptic operators and mixed derivatives

For a mixed derivative, the natural centered formula is

$$\partial_{xy} u(x_i, y_j) \approx \frac{U_{i+1,j+1} - U_{i+1,j-1} - U_{i-1,j+1} + U_{i-1,j-1}}{4h^2}.$$

Thus a general second-order operator produces a wider but still sparse stencil. A crucial warning is that consistency and ellipticity alone do not guarantee a discrete maximum principle. For example, discretizing

$$-a\,u_{xx} - 2b\,u_{xy} - c\,u_{yy}, \qquad a > 0, \quad ac - b^2 > 0,$$

with the formula above creates diagonal-neighbor coefficients of both signs. Monotonicity may require coefficient restrictions, a rotated stencil, or a wider stencil.

7.5 A compact fourth-order Poisson scheme

Higher order usually requires more grid points, but the Poisson equation has useful compact formulas. Define the nine-point Laplacian

$$(\Delta_9 U)_{ij} = \frac{1}{6h^2}\left[4 \sum_{\text{edge nbrs}} U + \sum_{\text{corner nbrs}} U - 20 U_{ij}\right].$$

For a smooth function,

$$\Delta_9 u = \Delta u + \frac{h^2}{12}\Delta^2 u + \mathcal{O}(h^4).$$

Since $-\Delta u = f$, a fourth-order compact discretization is

$$-\Delta_9 U = f + \frac{h^2}{12}\Delta f \quad \text{at interior nodes.} \tag{7.5}$$

The right-hand side correction is essential; applying the nine-point stencil with the unmodified right-hand side is generally only second order.

Three dimensions

The standard discretization of $-\Delta$ in three dimensions uses the center and its six coordinate neighbors. The maximum-principle argument, Kronecker-sum matrix representation, and consistency analysis extend directly, although the linear systems become much larger.

Exercises

Exercise 7.1 (Kronecker eigenpairs)**.** Use the one-dimensional eigenvectors to find every eigenvalue of the two-dimensional five-point matrix. Determine the asymptotic growth of its Euclidean condition number.

Exercise 7.2 (Rectangular grid)**.** Derive the five-point formula when $h_x \neq h_y$. Construct a quadratic barrier and obtain an explicit maximum-norm stability constant.

Exercise 7.3 (Nine-point expansion)**.** Expand all nine stencil values through sixth order and verify $\Delta_9 u = \Delta u + (h^2/12)\Delta^2 u + \mathcal{O}(h^4)$.

Exercise 7.4 (Anisotropy and monotonicity)**.** For constant a, b, c, write every coefficient in the centered stencil for $-au_{xx} - 2bu_{xy} - cu_{yy}$. Determine conditions under which all off-diagonal coefficients of the resulting matrix are nonpositive.

8 Nonuniform grids and irregular boundaries

8.1 The optimal three-point second derivative

Let

$$a = x_0 < x_1 < \cdots < x_N = b, \qquad h_i = x_i - x_{i-1}.$$

At x_i, the three-point approximation obtained by matching constants, linear functions, and quadratics is

$$\begin{aligned} D_h^2 U_i &= \frac{2}{h_i + h_{i+1}} \left(\frac{U_{i+1} - U_i}{h_{i+1}} - \frac{U_i - U_{i-1}}{h_i} \right) \\ &= \frac{2}{h_{i+1}(h_i + h_{i+1})} U_{i+1} - \frac{2}{h_i h_{i+1}} U_i + \frac{2}{h_i(h_i + h_{i+1})} U_{i-1}. \end{aligned} \tag{8.1}$$

For $-u'' = f$, set $-D_h^2 U_i = f(x_i)$. The off-diagonal coefficients of $-D_h^2$ are negative, so the discrete maximum principle survives on arbitrary positive step sizes.

8.2 What order is the nonuniform formula?

A Taylor expansion about x_i, with $h_- = h_i$ and $h_+ = h_{i+1}$, gives

$$D_h^2 u(x_i) = u''(x_i) + \frac{h_+ - h_-}{3} u^{(3)}(x_i) + \frac{h_+^2 - h_+ h_- + h_-^2}{12} u^{(4)}(x_i) + \mathcal{O}(h_{\max}^3). \tag{8.2}$$

A nonuniform three-point formula is not always second order

On an arbitrary quasi-uniform mesh, $h_+ - h_-$ can be of order $h_{\max}$, and (8.1) is then only first-order accurate locally. It is second order on a uniform mesh, or more generally on a smoothly graded mesh satisfying $|h_{i+1} - h_i| = \mathcal{O}(h_{\max}^2)$. A bounded mesh ratio alone does not cancel the cubic term.

This distinction is important: the stencil is monotone on any mesh, but its consistency order depends on how neighboring step sizes vary.

8.3 The Shortley–Weller idea

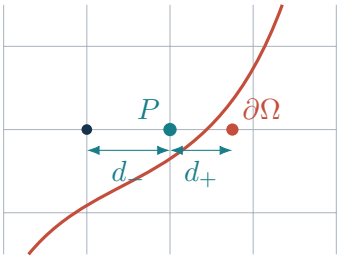
Let $\Omega \subset \mathbb{R}^2$ have a curved boundary and overlay it with a Cartesian grid. At a grid point $P \in \Omega$, a coordinate line may meet $\partial\Omega$ before reaching the next Cartesian node. Let d_- and d_+ be the distances from P to the available node or boundary intersection in the negative and positive x-directions. Use the nonuniform formula

$$D_{xx}^{SW} U(P) = \frac{2}{d_- + d_+} \left[\frac{U(P + d_+ e_x) - U(P)}{d_+} - \frac{U(P) - U(P - d_- e_x)}{d_-} \right], \tag{8.3}$$

inserting the prescribed value g when an endpoint is on the physical boundary. Define D_{yy}^{SW} analogously and solve

$$-(D_{xx}^{SW} + D_{yy}^{SW}) U(P) = f(P).$$

At regular nodes this is exactly the five-point method. At cut stencils, the local formula is generally only first order because the opposite distances are unequal.



8.4 Why first-order local error need not destroy global order

The low-order truncation error is confined to a thin set of nodes near the boundary. A generalized maximum-principle argument keeps separate track of interior and near-boundary residuals and uses different barrier functions for them. Under standard smoothness and geometric assumptions, the Shortley–Weller method can still satisfy a nodal estimate of the form

$$\|I_h u - U\|_{\ell^\infty(\Omega_h)} \le C_\Omega h^2 \|u\|_{C^4(\overline{\Omega})}.$$

The exact hypotheses matter; a local first-order statement alone neither proves nor disproves second-order global convergence.

An alternative is to use linear interpolation along a cut segment to eliminate a near-boundary unknown. Although linear interpolation sounds first order, its pointwise interpolation error for a C^2 function is $\mathcal{O}(h^2)$.

Monotonicity as a design principle

For elliptic problems, a stencil with positive diagonal, nonpositive off-diagonal coefficients, and suitable row sums often produces an M-matrix. This connects:

$$\text{discrete maximum principle} \iff \text{nonnegative inverse} \implies \text{maximum-norm stability}.$$

High order, compactness, and monotonicity can compete, especially for anisotropic operators and irregular meshes.

Exercises

Exercise 8.1 (Undetermined coefficients)**.** Starting from $\alpha U_{i+1} + \beta U_i + \gamma U_{i-1}$, impose exactness on 1, x, and x^2, and derive (8.1).

Exercise 8.2 (Local order)**.** Verify (8.2). Give one family of meshes for which the method is first order and one smoothly graded family for which it is second order.

Exercise 8.3 (Shortley–Weller coefficients)**.** Expand (8.3) into its three coefficients. Prove that the corresponding contribution to $-\Delta_h$ has nonpositive off-diagonal coefficients for every $d_-, d_+ > 0$.

Exercise 8.4 (Boundary interpolation)**.** Let x_b be a boundary intersection between two collinear points whose separation is $\mathcal{O}(h)$. Derive the remainder of linear interpolation at x_b and show that it is $\mathcal{O}(h^2)$ for $u \in C^2$.

9 Notation at a glance

Symbol	Meaning
$I_h u$	restriction or interpolation of u to a discrete space
δ_h^2	centered second difference on a uniform grid
$\mathcal{L}_h$	discrete differential operator
τ_h	truncation error or discrete residual of the exact solution
$\|\cdot\|_{\ell^\infty}$	maximum over the relevant grid nodes
H_0^1	H^1 functions with zero trace
H^{-1}	dual of H_0^1
V_h	conforming finite element subspace
$a(\cdot,\cdot)$	continuous bilinear form
A_h	algebraic matrix of a discrete problem

Suggested reading

For elliptic PDE theory and maximum principles, see [1]. Weak formulations and finite elements are developed in [2, 3]. Broad treatments of finite differences, boundary closures, and numerical PDEs appear in [4, 5, 6].

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